Subject: تمرین اول تحلیل هوشمند تصاویر پزشکی
Year. Month. Date. ( ) 402212214 سید امیر کسائی

a) if $y[n] = x[n] * h[n]$, then $y[n-1] = x[n-1] * h[n-1]$ (1)

$\Rightarrow$ based on time shift property in convolution:

$$y[n-n_0] = x[n-n_0] * h[n] \quad \text{or} \quad y[n-n_0] = x[n] * h[n-n_0]$$

$$\Rightarrow y[n-n_0-m_0] = x[n-n_0] * h[n-m_0]$$

$$\rightarrow x[n-1] * h[n-1] = y[n-1-1] = y[n-2] \neq y[n-1] \Rightarrow \text{False}$$

b) if $y(t) = x(t) * h(t)$, then $y(-t) = x(-t) * h(-t)$

$\Rightarrow$ based on time scaling property in convolutions:

$$x(\alpha t) * h(\alpha t) = \frac{y(\alpha t)}{|\alpha|}$$

$$\xrightarrow{\alpha=-1} x(-t) * h(-t) = \frac{1}{|-1|} y(-t) \Rightarrow x(-t) * h(-t) = y(-t) \Rightarrow \text{True}$$

(2)

LTI عمومی

$$\begin{cases} S1: w[n] = \frac{1}{2} w[n-1] + x[n] \\ S2: y[n] = \alpha y[n-1] + \beta w[n] \end{cases}, \quad y[n] = -\frac{1}{8} y[n-2] + \frac{3}{4} y[n-1] + x[n]$$

a) we can use Z transform:

$$S1 \xrightarrow{Z} W(z) = \frac{1}{2} z^{-1} W(z) + X(z) \rightarrow W(z) = \frac{X(z)}{1-\frac{z^{-1}}{2}} \quad (1)$$

$$S2 \xrightarrow{Z} Y(z) = \alpha z^{-1} Y(z) + \beta W(z) \quad (2)$$

$$\xrightarrow[(2)]{(1)} Y(z) = \alpha z^{-1} Y(z) + \beta \left(\frac{X(z)}{1-\frac{z^{-1}}{2}} \right) \rightarrow (1-\alpha z^{-1}) Y(z) = \beta \left(\frac{X(z)}{1-\frac{z^{-1}}{2}} \right)$$

Subject:
Year. Month. Date. ( )

$$\rightarrow Y(z)\left(1-\alpha z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)=\beta X(z)$$

$$\rightarrow Y(z)\left(1-\frac{z^{-1}}{2}-\alpha z^{-1}+\frac{\alpha z^{-2}}{2}\right)=\beta X(z)$$

$$\rightarrow Y(z)\left(1-\left(\frac{1}{2}+\alpha\right)z^{-1}+\frac{\alpha z^{-2}}{2}\right)=\beta X(z) \quad (3)$$

$$y[n]=-\frac{1}{8}y[n-2]+\frac{3}{4}y[n-1]+x[n] \xrightarrow{z} Y(z)=-\frac{1}{8}z^{-2}Y(z)+\frac{3}{4}z^{-1}Y(z)+X(z)$$

$$\rightarrow Y(z)\left(\frac{1}{8}z^{-2}-\frac{3}{4}z^{-1}+1\right)=X(z) \quad (4)$$

$$\xrightarrow[(4)]{(3)} \beta=1, \qquad \frac{\alpha}{2}=\frac{1}{8}\rightarrow \alpha=\frac{1}{4}$$

b)

$$\rightarrow Y(z)=X(z)\cdot H(z) \rightarrow H(z)=\frac{Y(z)}{X(z)} \quad (5)$$

$$\xrightarrow{(4)} Y(z)\left(\frac{8z^2+1-6z}{8z^2}\right)=X(z) \rightarrow \frac{Y(z)}{X(z)}=\frac{8z^2}{8z^2+1-6z} \quad (6)$$

$$\xrightarrow[(6)]{(5)} H(z)=\frac{8z^2}{8z^2+1-6z}=\frac{z^2}{z^2+\frac{1}{8}-\frac{6z}{8}}=\frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}$$

$$\rightarrow \frac{H(z)}{z}=\frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)} \quad (7) \quad =\frac{A}{z-\frac{1}{2}}+\frac{B}{z-\frac{1}{4}}$$

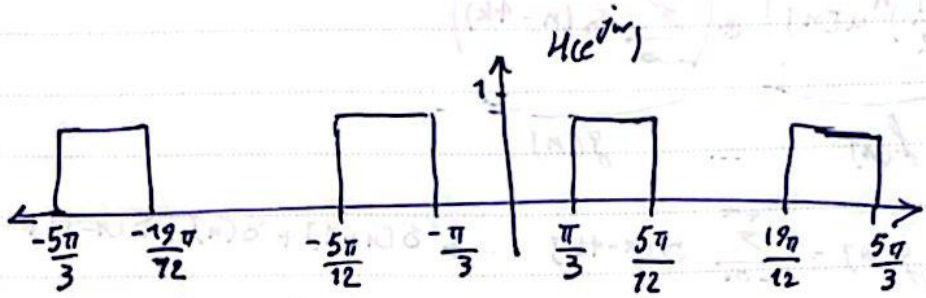
$$\rightarrow =\frac{A\left(z-\frac{1}{4}\right)+B\left(z-\frac{1}{2}\right)}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)} \quad (8)$$

$$\xrightarrow[(8)]{(7)} A\left(z-\frac{1}{4}\right)+B\left(z-\frac{1}{2}\right)=z \rightarrow (A+B)z-\frac{A}{4}-\frac{B}{2}=z$$

$$\rightarrow \begin{cases} A+B=1 \\ -\frac{A}{4}-\frac{B}{2}=0 \rightarrow -\frac{A}{2}=B \end{cases} \Rightarrow A=2,\ B=-1$$

$$\rightarrow \frac{H(z)}{z} = \frac{2}{z-\frac{1}{2}} - \frac{1}{z-\frac{1}{4}} \rightarrow H(z) = \frac{2z}{z-\frac{1}{2}} - \frac{z}{z-\frac{1}{4}}$$

$$\xrightarrow{\text{Inverse } z} h[n] = \left(2\left(\tfrac{1}{2}\right)^n - \left(\tfrac{1}{4}\right)^n\right) u[n]$$



3

a) $x_1[n] = 1 + \sin\left(\overbrace{\frac{3\pi}{8}}^{\omega} n + \frac{\pi}{4}\right)$

$$\omega = \frac{3\pi}{8} = \frac{2\pi}{N} \times m \rightarrow N = \frac{2\pi}{\omega} \times m = \frac{2\cancel{\pi}}{\frac{3\cancel{\pi}}{8}} \times m = \frac{16}{3} \times m$$

min int that makes N an int

for $m=3 \rightarrow N = 16$

$$\rightarrow x_1[n] = 1 + \sin\left(3\left(\tfrac{2\pi}{16}\right)n + \tfrac{\pi}{4}\right)$$

$$\xrightarrow{\text{Fourier Series}} = \underbrace{1}_{a_0} + \underbrace{\frac{1}{2j}\left(e^{j\frac{\pi}{4}} \cdot e^{3\left(\frac{2\pi}{16}\right)n}\right)}_{a_3} - \underbrace{\frac{1}{2j}\left(e^{-j\frac{\pi}{4}} \cdot e^{-3\left(\frac{2\pi}{16}\right)n}\right)}_{a_{-3}}$$

in range $[0,15]$ we do not have -3. so we can change it based on.

signals periodic pattern $\Rightarrow a_k = a_{k+N} \xrightarrow{N=16} a_{-3} = a_{13}$

$$\rightarrow x_1[n] = 1 + \frac{1}{2j}\left(e^{j\frac{\pi}{4}} e^{3\left(\frac{2\pi}{16}\right)n}\right) + \frac{-1}{2j}\left(e^{13\left(\frac{2\pi}{16}\right)} e^{-j\frac{\pi}{4}}\right)$$

$$y_1[n] = \sum_{k=0}^{N-1=15} a_k H\left(e^{j\frac{2\pi}{16}k}\right) \cdot \left(e^{j\frac{2\pi}{16}kn}\right)$$

$$= 1 \cdot H(0) + \left(\frac{1}{2j} e^{j\frac{\pi}{4}}\right) H\left(e^{3\left(\frac{2\pi}{16}\right)n}\right) e^{j\frac{3\pi}{8}n} - \left(\frac{1}{2j} e^{-j\frac{\pi}{4}}\right) H\left(e^{13\left(\frac{2\pi}{16}\right)n}\right) e^{j\frac{\pi}{8}(13n)}$$

Subject:
Year. Month. Date. ( )

$$= \frac{1}{2j} e^{j(\frac{3\pi}{8}+\frac{\pi}{4})} - \frac{1}{2j} e^{-j(\frac{3\pi}{8}n+\frac{\pi}{4})} = \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$$

b) $x_2[n] = \sum_{k=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{n-4k} u[n-4k]$

we can convert signal to a convolution with .. pulse signal

$$\rightarrow x_2[n] = \underbrace{\left(\left(\frac{1}{2}\right)^n u[n]\right)}_{f[n]} * \underbrace{\left(\sum_{-\infty}^{+\infty} \delta(n-4k)\right)}_{g[n]}$$

$$\Rightarrow g[n] = \sum_{-\infty}^{+\infty} \delta[n-4k] = \dots + \delta[n+4] + \delta[n] + \delta[n-4] + \dots$$

$\rightarrow$ period of $g[n]$ is $\underline{\underline{N=4}} \rightarrow \omega_0 = \frac{2\pi}{N} = \frac{\pi}{2}$

we know that:

$$a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\omega_0 kn} \xrightarrow{\text{for } g} a_k = \frac{1}{N}\sum_{n=0}^{3} g[n] e^{-j\frac{\pi}{2}kn}$$

$$\rightarrow a_k = \frac{1}{4}\left(g[0] \cdot e^{-j\frac{k\pi}{2}(0)} + g[1] e^{-j\frac{k\pi}{2}(1)} + g[2] \cdot e^{-j\frac{k\pi}{2}(2)} + g[3] \cdot e^{-j\frac{k\pi}{2}(3)}\right)$$

for $g[n]$, we have $N=4 \rightarrow g[1], g[2], g[3] = 0$

$$\Rightarrow a_k = \frac{1}{4}\overbrace{g[0]}^{1} = \underbrace{\frac{1}{4}}_{a_0}$$, we know that $\rightarrow g[n] = \sum_{-\infty}^{+\infty} a_k H(e^{jk\omega}) e^{jk\omega n}$

$\rightarrow$ we pass $g[n]$ through filter, so the out put $r[n]$ would be:

$$r[n] = \sum_{k=-\infty}^{+\infty} a_k H(e^{j\frac{k\pi}{2}}) e^{jk\frac{\pi}{2}n} = \sum_{k=1}^{3} a_k H(e^{j\frac{k\pi}{2}}) e^{jk\frac{\pi}{2}n}$$

for all k, except $k=0 \rightarrow a_k = 0 \rightarrow r[n] = a_0 \underbrace{H(e^{j\frac{\pi}{2}(0)})}_{0} e^{j\frac{\pi}{2}(0)n}$

$\Rightarrow r[n] = \frac{1}{4}(0) = 0$, now we should convolve $r[n]$ with $f[n]$ to find the output

as we know the $f[n]$ signal is not periodic. So it does not change the frequency.

$\Rightarrow y_2[n] = f[n] * r[n] = 0 \rightarrow y_2[n] = 0$

Subject:
Year. Month. Date. ( )

a) lets suppose that second image is a filter. size of second image 4

so for each point in the first image we have to mak N^2 multiplications and N^2 addition. so the time complexity for each point would be N^2. the image is $N \times N$, so the whole computation completes in $O(N^4)$.

b) we can solve this problem in frequency domain.

for converting to frequency domain we can use Fast Fourier Transform which transforms a 1D to fourier domain in $O(N \log N)$

for a 2D image it takes $O(N^2 \log N)$

so for computing the convolution first we transform images into fourier domain using FFT in $O(2N^2 \log N) = O(N^2 \log N)$. then we do the multiplication in frequency domain in $O(N)^2$, and at the end we transform the output back to original domain which also takes $O(N^2 \log N)$

So overall the computation take $O(N^2 \log N)$